StayInsight

ACCOMMODATION INTELLIGENCE

Paris — Airbnb Analysis

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IronHack Data Analytics Bootcamp

Project Overview

Why this topic?

- StayInsight is a travel intelligence startup helping accommodation services make smarter strategic decisions in European cities
- Paris chosen as first city — richer data availability across hospitality, especially Airbnb
- Goal: understand what drives pricing, satisfaction and location patterns

Datasets

- Airbnb Paris weekdays & weekends (Kaggle)
- Hotel data via Google Hotel Scraper API (RapidAPI)
- POI enrichment via Geoapify Places API
- Tourism & Hospitality macro dataset (Kaggle)

Research Questions & Hypotheses

RQ1 — Weekday vs Weekend

H: Airbnb listings will have lower cleanliness ratings and guest satisfaction scores on weekends due to higher turnover and less time for cleaning between stays.

RQ2 — Superhost Impact

H: Superhost listings will show higher guest satisfaction scores and cleanliness ratings than non-superhost listings, and will command higher prices per night.

RQ3 — Location, Satisfaction & Pricing

H: Listings in areas with higher tourism intensity — rich in heritage and major attractions — tend to command higher prices, but do not necessarily achieve higher guest satisfaction.

Data Wrangling & Cleaning

Missing data & duplicates

Nulls handled per column with justified strategy. Duplicates checked after each merge and drop.

Formatting & standardisation

All string columns stripped and title-cased. Address lines cleaned of list formatting. City names standardised.

API — Google Hotels

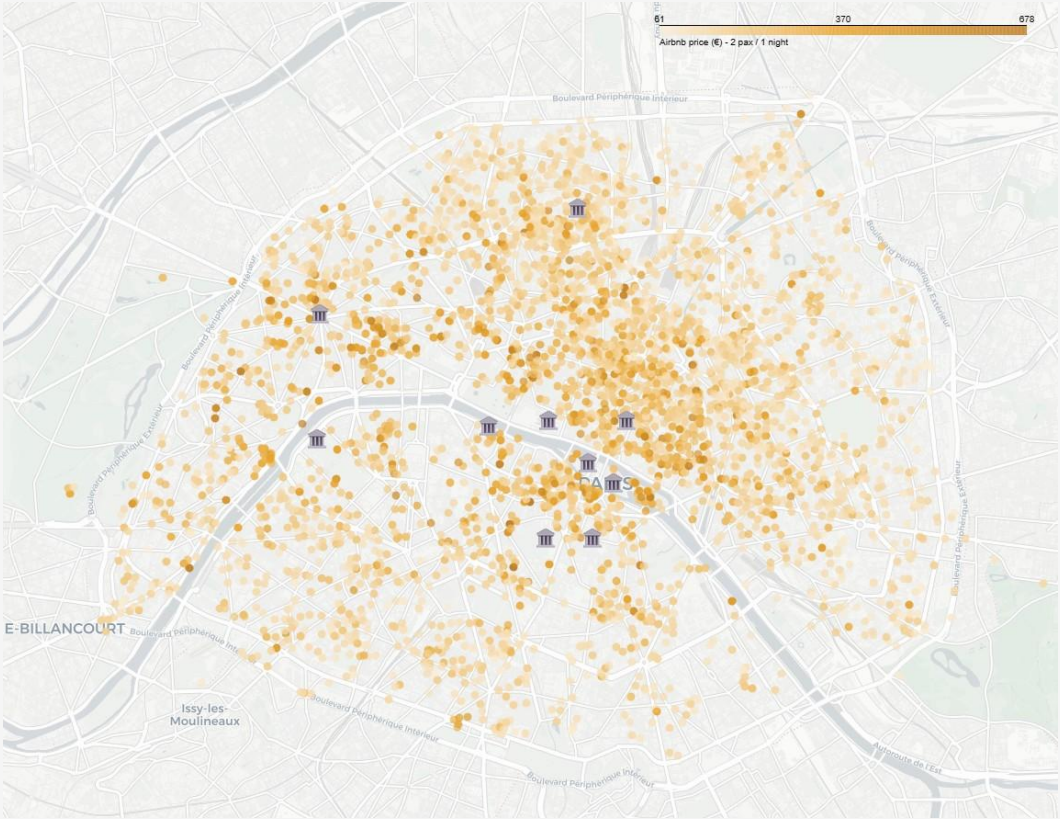
Rome rows dropped. Hotels beyond 50km from Paris centre removed using Geopy distance calculation on lat/lon.

API — Geoapify POI

Extracted POIs within 500m radius. Derived n_geoapify_heritage and geoapify_tourism_density_score.

Exploratory Data Analysis

- Uncovered patterns linking cleanliness, guest satisfaction, and pricing across listings.
- Mapped Airbnb listings across Paris neighbourhoods.
- Investigated whether tourism intensity (POIs nearby) influences pricing and satisfaction.



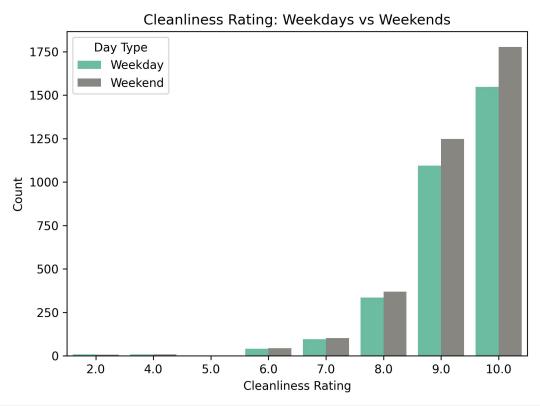
Conclusions & Insights

RQ1 — Weekday vs Weekend

H: Lower cleanliness & satisfaction on weekends due to higher turnover

Conclusion: Weekends actually have slightly better satisfaction, cleanliness, and higher prices

	Average
Weekday	91,85
Weekend	92,20



	Average
Weekday	9,25
Weekend	9,27

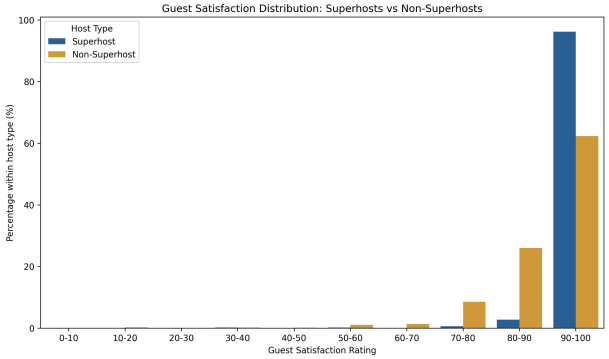
	Average
Weekday	193,51
Weekend	199,39

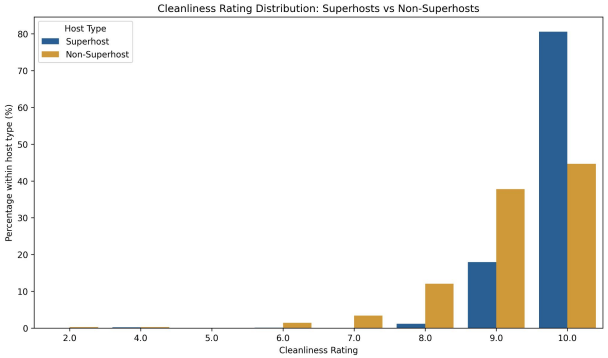
Conclusions & Insights

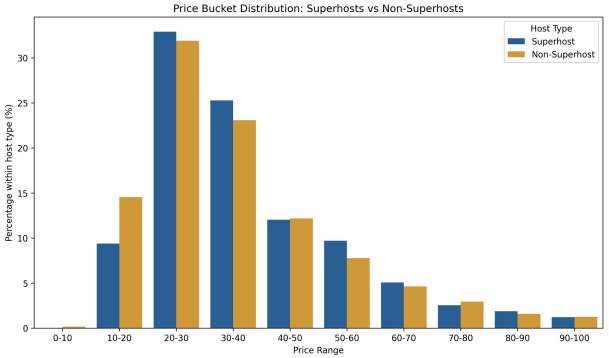
RQ2 — Superhost Impact

H: Superhosts show higher satisfaction, cleanliness and prices

Conclusion: Superhosts have higher cleanliness ratings and satisfactions scores







	Average
SuperHost	97
Not SuperHost	91

	Average
SuperHost	9.78
Not SuperHost	9.18

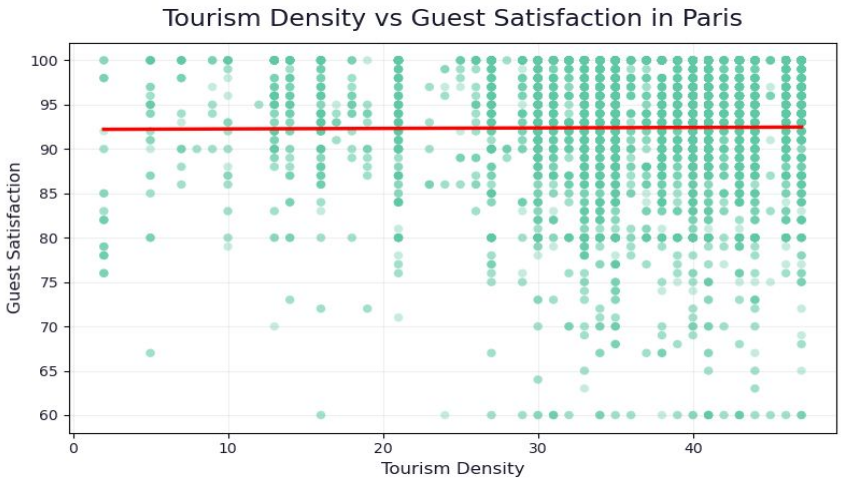
	Median
SuperHost	170.45
Not SuperHost	156.47

Conclusions & Insights

RQ3 — Location & Pricing

H: Higher tourism intensity = higher prices, higher satisfaction

Conclusion: Location influences prices, but not guest satisfaction



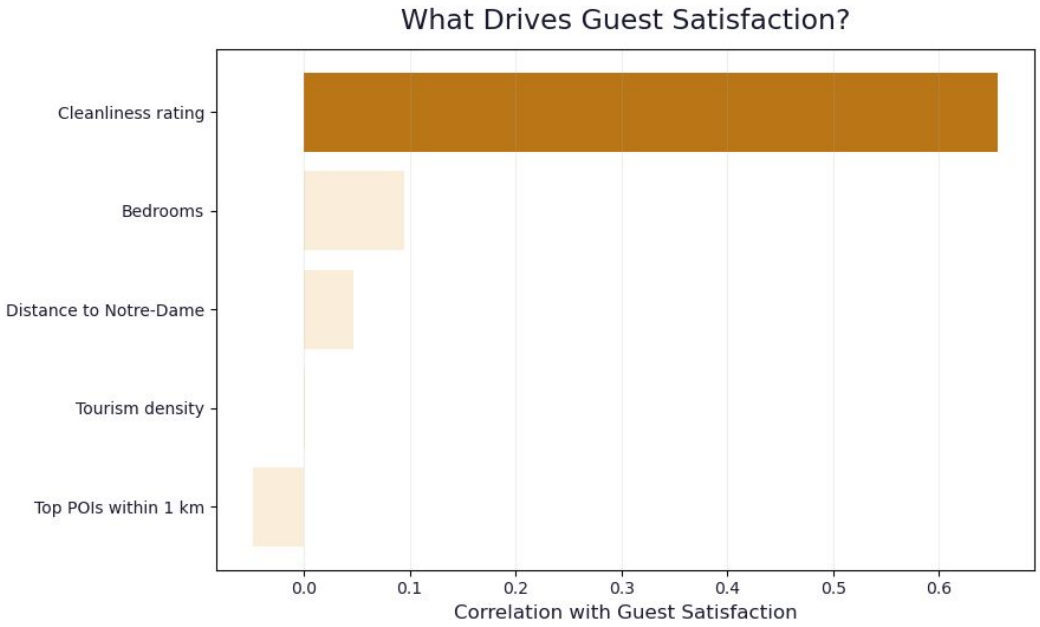
Digging deeper: What actually drives guest satisfaction?

Key findings

- Cleanliness is the strongest driver of satisfaction
- Facility quality and comfort also play a key role
- Tourism-related factors have minimal impact

Conclusion

Guest experience depends on internal quality, not location
What drives price ≠ what drives satisfaction



Major Obstacle

The obstacle

Finding a suitable dataset was our biggest challenge. Available data didn't align with our initial research questions.

What we learned

Defining questions and searching for datasets is an iterative process: if no dataset fits, pivot and keep looking until scope and data align.

How it influenced us

Gave us practical knowledge on how to handle data gaps in future projects. In the workplace, we'll proactively ask for more data and seek support from others: just as we did with Miguel.

What we'd do differently

Get a proper API budget — data quality scales with the number of requests you are allowed to make.

Teamwork & Project Management

Workflow

Used Trello Kanban board to track tasks. Each member owned specific cards with checklists.

What worked well

- Kanban board for task tracking
- Regular check-ins every 1h–1h30
- Strong team support and collaboration
- Clear division of work per question/hypothesis

What to improve

- Team communication outside of check-ins to ensure everyone stays on the same page

Risk management

- Difficulty finding a suitable dataset
- Had to pivot from the original plan
- Found an API and alternative dataset, adapted questions to the new scope

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Thank You!

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